Schur leading terms and a toric degeneration of the Lecture Hall algebra

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October 13, 2026

Abstract

Starting from the signed-moment identification of the Lecture Hall polynomials with signed Stieltjes–Rogers moments, we determine the leading term of every proposed SAGBI generator. The key identity is a highest-degree recursion for Schur determinants of arbitrary Stieltjes–Rogers moments. Its iteration reads off the Frobenius coordinates of the indexing partition. After the Lecture Hall sign specialization, every leading coefficient is 1, and the resulting exponents give an explicit bijection from subsets of $[n-1]$ to the Hilbert basis of the Lecture Hall cone.

We then prove that the initial algebra is the entire Lecture Hall semigroup ring, for every n and over every field. The last step uses equality of elementary lattice-point growth constants, not the Lecture Hall Theorem. Thus the argument proves Katthän's Conjecture 4.2 and answers Problem 5.1, with the explicitly stated square-minor interpretation. It also produces a monic integral SAGBI basis and an explicit free, hence flat, one-parameter degeneration. All needed determinant and counting arguments are included. Finite exact records and code accompany this note; they are not used as proofs. No claim of publication priority is made.

1 Definitions and statements

All variables below are algebraically independent unless a specialization is explicitly stated. Let $\mu_j(\alpha_1, \alpha_2, \ldots)$ be the sum of weights of Dyck paths of semilength j: up steps have weight 1 and a down step from height h has weight α_h. Set $\mu_0 = 1$ and $\mu_j = 0$ for $j < 0$. Define

$$\epsilon_h = \begin{cases} -1 & h \geq 4 \text{ is even,} \\ 1 & \text{otherwise,} \end{cases} \qquad \alpha_h = \epsilon_h y_h y_{h+1}, \qquad \ell_{j+1} = y_1 \mu_j(\alpha) \quad (j \geq 0). \tag{1}$$

Also set $\ell_j = 0$ for $j \leq 0$. The polynomiality note [2] establishes that these are the Lecture Hall polynomials. Appendix A reproves the needed identification so that the present argument is self-contained.

For $S = \{s_1 < \cdots < s_r\} \subseteq [n-1]$, $r > 0$, put

$$\ell_S = (-1)^{r+1} \det[\ell_{s_j+2-i}]_{1 \leq i,j \leq r}, \qquad \ell_\varnothing = \ell_1 = y_1. \tag{2}$$

This is the negative of the top-row minor of the negative upper-triangular Toeplitz matrix in Katthän's Definition 4.1 [1]. Let

$$A_n(k) = k[\ell_1, \ldots, \ell_n] \subseteq k[y_1, \ldots, y_n], \qquad \mathcal{B}_n = \{\ell_S : S \subseteq [n-1]\}.$$

The term order is degree-lexicographic, with $y_1 > \cdots > y_n$. For a polynomial its leading term includes its coefficient. Write

$$C_n = \left\{ x \in \mathbb{R}^n : \frac{x_1}{n} \geq \frac{x_2}{n-1} \geq \cdots \geq x_n \geq 0 \right\}, \qquad L_n = C_n \cap \mathbb{Z}^n. \tag{3}$$

The semigroup algebra $k[L_n]$ is the subalgebra spanned by the monomials y^e with $e \in L_n$.

Theorem 1.1 (All dimensions; all characteristics)**.** *For every $n \geq 1$ and every field k,*

$$\mathrm{in}_{\mathrm{deglex}} A_n(k) = k[L_n].$$

The set $\mathcal{B}_n$ is a minimal SAGBI basis. Its leading coefficients are all 1*, and its leading exponents run bijectively through*

$$\mathcal{H}_n = \{(1,0,\ldots,0)\} \cup \{(a+1,a,b_3,\ldots,b_t,0,\ldots,0) : 1 \leq a \leq n-1, \quad a > b_3 > \cdots > b_t > 0\}, \tag{4}$$

where the tail after a may be empty. This set is the Hilbert basis of L_n.

Here is the promised explicit formula. For nonempty S define a partition of length r by

$$\lambda_i = s_{r+1-i} - r + i \qquad (1 \leq i \leq r). \tag{5}$$

Let D be its Durfee size, and let

$$A_i = \lambda_i - i, \qquad B_i = \lambda'_i - i \quad (1 \leq i \leq D)$$

be its Frobenius arm and leg lengths. Form the sequence

$$(d_1, d_2, d_3, d_4, \ldots) = (A_1 + 1, B_1, A_2 + 1, B_2, \ldots, A_D + 1, B_D), \tag{6}$$

remove a final zero if present, and pad with zeros thereafter.

Theorem 1.2 (Leading-term formula and the subset map)**.** *For every nonempty $S \subseteq [n-1]$,*

$$\mathrm{LT}(\ell_S) = y_1^{r+d_1} \prod_{j=2}^{n} y_j^{d_{j-1}+d_j}. \tag{7}$$

In particular the first two exponents are $\max(S)+1$ *and* $\max(S)$*. Together with* $\mathrm{LT}(\ell_\varnothing) = y_1$*, formula* (7) *is an explicit, invertible description of the map in Katthän's Problem 5.1.*

The source conventions matter. The square minor defining E_i uses rows $1, \ldots, \lceil i/2 \rceil$ and columns $\lfloor i/2 \rfloor + 1, \ldots, i$. This is the interpretation in [2], supported by the source's examples and Definition 4.1. We likewise retain the corrected fourth polynomial consistent with the defining equations and the source's Section 6. These are not represented as author-issued errata.

2 A determinant identity from reciprocal series

We first work with an arbitrary sequence $h_0 = 1, h_1, h_2, \ldots$ over a commutative ring, with $h_j = 0$ for $j < 0$. Define e_j by

$$\left(\sum_{j \geq 0} h_j z^j\right)\left(\sum_{j \geq 0} (-1)^j e_j z^j\right) = 1, \qquad e_0 = 1,$$

and set $e_j = 0$ for $j < 0$. For a partition λ of length r, write

$$s_\lambda(h) = \det[h_{\lambda_i - i + j}]_{1 \leq i,j \leq r}, \qquad s_\varnothing(h) = 1.$$

Trailing zero parts do not change this determinant: the bottom row then has a 1 in its last position and zeros before it, and expansion removes that row and column.

Lemma 2.1 (Dual Jacobi–Trudi identity)**.** *For every partition λ,*

$$s_\lambda(h) = s_{\lambda'}(e).$$

Proof. The empty case is immediate. Set $L = \lambda_1$, $N = r + L$, and index the $N \times N$ matrix U by $0, \dots, N-1$, with $U_{ab} = h_{a-b}$. It is lower triangular with determinant 1, and its inverse has entries $(U^{-1})_{ab} = (-1)^{a-b} e_{a-b}$, by the reciprocal-series identity.

Use the row set $I = \{\lambda_i + r - i : 1 \le i \le r\}$ and column set $J = \{0, \dots, r-1\}$. Reversing both ordered index sets identifies the minor $U_{I,J}$ with $s_\lambda(h)$. The complement of I is, in increasing order,

$$I^c = \{r + j - 1 - \lambda'_j : 1 \le j \le L\}, \qquad J^c = \{r, \dots, r + L - 1\}.$$

To check the first formula, its entries are strictly increasing, lie in $[0, N-1]$, and cannot equal $\lambda_i + r - i$: equality would say $\lambda_i + \lambda'_j = i + j - 1$, whereas if $\lambda_i \ge j$ the left side is at least $i + j$, and otherwise it is at most $i + j - 2$. The two sets therefore are complementary by cardinality.

The complementary-minor identity gives

$$\det U_{I,J} = (-1)^{\sum I + \sum J} \det(U^{-1})_{J^c, I^c}.$$

Here $\sum I + \sum J \equiv |\lambda| \pmod 2$. The transposed complementary minor has entries

$$(-1)^{\lambda'_j - j + i} e_{\lambda'_j - j + i}.$$

Their determinant has the overall sign $(-1)^{|\lambda|}$, so the two signs cancel.

For completeness, the complementary-minor identity follows by permuting selected rows and columns to the front and writing the resulting matrix in four blocks A, B, C, D, in row order. When A is invertible, the determinant of the whole matrix is $\det A \det(D - CA^{-1}B)$, while the corresponding bottom-right block of the inverse is $(D - CA^{-1}B)^{-1}$. The permutation signs give the displayed formula. Proving this first for a generic matrix and clearing denominators makes it a polynomial identity, hence valid for the present matrix as well. □

Return now to the Dyck moments with independent weights α_h. Write $h_j = \mu_j(\alpha_1, \alpha_2, \dots)$ and $\widehat{h}_j = \mu_j(\alpha_2, \alpha_3, \dots)$. First-return decomposition of a Dyck path gives the formal identity

$$H(z) := \sum_{j \ge 0} h_j z^j = \frac{1}{1 - \alpha_1 z \widehat{H}(z)}, \qquad \widehat{H}(z) = \sum_{j \ge 0} \widehat{h}_j z^j. \tag{8}$$

Consequently

$$e_j = (-1)^{j-1} \alpha_1 \widehat{h}_{j-1} \qquad (j \ge 1). \tag{9}$$

All these are identities of formal power series; a coefficient of order j uses only finitely many heights.

Lemma 2.2 (Highest-degree recursion)**.** *For nonempty λ, set $L = \lambda_1$ and $\nu = (\lambda_2, \lambda_3, \dots)'$. Then*

$$\deg_{\alpha_1} s_\lambda(\mu) = L, \tag{10}$$

$$[\alpha_1^L] s_\lambda(\mu) = (-1)^{|\lambda| - L} s_\nu(\widehat{\mu}). \tag{11}$$

Proof. By Lemma 2.1, $s_\lambda(h)$ is the size-L determinant $\det[e_{\lambda'_i - i + j}]$. Every entry has degree at most 1 in α_1, by (9). To extract the coefficient of α_1^L, replace an entry of index $q \ge 1$ by $(-1)^{q-1} \widehat{h}_{q-1}$, and an entry of index $q \le 0$ by zero. Every determinant term has the same sign $(-1)^{|\lambda| - L}$. After removing this sign the determinant is

$$\det[\widehat{h}_{\lambda'_i - 1 - i + j}]_{1 \le i,j \le L} = s_{(\lambda'_1 - 1, \dots, \lambda'_L - 1)}(\widehat{h}) = s_\nu(\widehat{h}).$$

An entry of index $q \le 0$ is correctly zero because its new moment index $q - 1$ is negative. Trailing zero parts can be removed as noted above. Finally $|\nu| = |\lambda| - L < |\lambda|$. Induction on partition size, starting with $s_\varnothing = 1$, proves that the extracted coefficient is nonzero, and therefore the degree bound is exact. □

3 Frobenius coordinates give the generic leading term

Apply repeatedly the operation

$$\lambda \longmapsto (\lambda_2, \lambda_3, \ldots)'.$$

At each stage record the first row length. Denote the resulting positive sequence by $(d_1, \ldots, d_t)$.

Proposition 3.1 (Generic Schur–moment leading term)**.** *In lexicographic order* $\alpha_1 > \alpha_2 > \cdots$*,*

$$\operatorname{LT} s_\lambda(\mu(\alpha)) = (-1)^{\sum_{j\geq 1} d_{2j}} \prod_{h\geq 1} \alpha_h^{d_h}. \tag{12}$$

The d_h *are exactly the sequence in* (6)*, with a final zero omitted. In particular, every such generic Schur determinant is nonzero and has leading coefficient* 1 *or* -1*.*

Proof. Lemma 2.2 successively determines the greatest possible exponent of α_1, then of α_2 among those terms, and so on. It removes d_h cells at stage h, so the total sign exponent is

$$\sum_{h\geq 1} \sum_{j>h} d_j = \sum_{j\geq 1} (j-1) d_j \equiv \sum_{j\geq 1} d_{2j} \pmod 2.$$

For the coordinate formula, after two operations the remaining shape is $(\lambda_2 - 1, \lambda_3 - 1, \ldots)$, with zero parts removed. Thus after $2i-2$ operations, for $i \leq D$, its row lengths are the positive entries of $(\lambda_i - i + 1, \lambda_{i+1} - i + 1, \ldots)$; nonpositive entries are omitted. The next row length is $\lambda_i - i + 1 = A_i + 1$, and the row length after transposition and removal is $\lambda'_i - i = B_i$. The operations stop exactly when the shape is empty. This proves both claims. □

Remark 3.2. The proof uses only reciprocal-series algebra and determinant identities. It does not use positivity or analytic moment theory. The continued fraction/path framework is classical [4, 3]; the precise leading-term statement above is proved here, without a priority claim.

4 Specialization, signs, and the explicit bijection

Proof of Theorem 1.2. Substitute $\ell_{j+1} = y_1 \mu_j$ in (2), transpose the moment matrix, and reverse both its rows and columns. The two reversal signs cancel, and (5) gives

$$\ell_S = (-1)^{r+1} y_1^r s_\lambda(\mu(\alpha)). \tag{13}$$

This is valid also when the original Toeplitz minor has structural zeros, because negative moment indices are defined to be zero.

Under $\alpha_h = \epsilon_h y_h y_{h+1}$ a profile d becomes the exponent vector

$$(r + d_1, d_1 + d_2, d_2 + d_3, \ldots).$$

This map on profiles is injective and preserves lexicographic comparisons: at the first differing profile coordinate the corresponding y-exponents differ in the same direction. Thus distinct monomials do not merge. Moreover the Schur determinant is homogeneous of degree $|\lambda|$ in α, so all its specialized terms have total degree $r + 2|\lambda|$. Lexicographic and degree-lexicographic leading terms therefore coincide for this polynomial.

The generic sign from (12) is $(-1)^{\sum_i B_i}$. The specialized negative weights contribute $(-1)^{\sum_{i\geq 2} B_i}$. Combining these with the prefactor in (13), the final sign is $(-1)^{r+1+B_1} = 1$, because $B_1 = r - 1$. This proves (7). Since $d_1 = \lambda_1 = \max(S) - r + 1$ and $d_2 = r - 1$, the first two exponents have the asserted values. □

Proposition 4.1 (Image and inverse)**.** *Formula* (7)*, together with the empty subset, gives a bijection onto the set* $\mathcal{H}_n$ *in* (4)*.*

Proof. For a nonempty partition its positive profile is consecutive, and

$$d_i > d_{i+2} \quad \text{whenever } d_i > 0,$$

where missing terms are zero. This follows from the strict decrease of the Frobenius arm lengths and of the leg lengths. If the leading exponent is $e = (r + d_1, d_1 + d_2, \ldots)$, then $e_1 = e_2 + 1$, and

$$e_j - e_{j+1} = d_{j-1} - d_{j+1} > 0$$

through its positive tail, for $j \geq 2$. Also $e_2 = \max(S) \leq n - 1$. Since $e_2 \leq n - 1$, its strictly decreasing positive tail has at most $n-1$ entries, so the vector has at most n nonzero coordinates. Thus $e \in \mathcal{H}_n$.

Here is a constructive inverse. Omit trailing zeros from $e \in \mathcal{H}_n$. For $e = (1)$ use $S = \varnothing$. Otherwise put

$$d_i = e_{i+1} - e_{i+2} + e_{i+3} - \cdots \quad (1 \leq i < \text{length}(e)). \tag{14}$$

All these numbers are positive, since the tail is strictly decreasing. Also $d_i - d_{i+2} = e_{i+1} - e_{i+2} > 0$. The two sequences

$$A_i = d_{2i-1} - 1, \qquad B_i = d_{2i},$$

with a final $B_i = 0$ when needed, are strictly decreasing nonnegative integer sequences of the same length. They define a unique partition: for $1 \leq i \leq D$ its row length is $i + A_i$, and for $i > D$ its row length is $\#\{j : j + B_j \geq i\}$. To verify this construction, place a diagonal cell at (i, i), its A_i cells to the right, and its B_i cells below, for each i. Strict decrease ensures that the union is a Ferrers diagram and recovers exactly these arm and leg lengths.

Let λ be this partition and $r = \ell(\lambda) = B_1 + 1 = d_2 + 1$, where $d_2 = 0$ if absent. Then set

$$S = \{\lambda_i + r - i : 1 \leq i \leq r\}. \tag{15}$$

Its elements are distinct positive integers and its largest is $d_1 + r - 1 = e_1 - 1 \leq n - 1$. Equations (14) give $e_j = d_{j-1} + d_j$ for $j \geq 2$, and $r + d_1 = e_1$ because $e_1 = e_2 + 1$. Hence this inverse really recovers e and the original S. □

For example, $S = \{2, 5, 7\}$ gives $\lambda = (5, 4, 2)$ and $d = (5, 2, 3, 1)$. Therefore

$$\text{LT}(\ell_{\{2,5,7\}}) = y_1^8 y_2^7 y_3^5 y_4^4 y_5.$$

Conversely, alternating tails of $(8, 7, 5, 4, 1)$ recover $(5, 2, 3, 1)$. The rank r can also be recovered directly as $e_1 - e_2 + e_3 - e_4 + \cdots$.

5 The Hilbert basis and an equal-growth principle

Lemma 5.1 (Elementary Hilbert-basis description)**.** *The indecomposable nonzero elements of L_n are precisely $\mathcal{H}_n$. Every $x \in L_n$ is a sum of elements of $\mathcal{H}_n$.*

Proof. For $n = 1$ the assertion is immediate. Suppose $n \geq 2$ and put $g = x_1 - x_2$. The first normalized inequality implies $x_2 \leq (n-1)g$. Hence $g \geq 1$ if $x \neq 0$. The integer points with $g = 1$ are exactly (4): $x_2 \leq n - 1$, and any two consecutive positive coordinates must decrease strictly. Conversely, an integer vector of the displayed form belongs to C_n. Indeed $x_{i+1} \leq n - i$, and whenever it is positive, $x_i - x_{i+1} \geq 1$; therefore $(n-i)(x_i - x_{i+1}) \geq x_{i+1}$, which is the normalized inequality.

For a general nonzero x construct g vectors, indexed by $t = 0, \ldots, g - 1$. For $i \geq 2$ put

$$b_i^{(t)} = \min\{n - i + 1, \max\{0, x_i - t(n - i + 1)\}\}, \qquad b_1^{(t)} = b_2^{(t)} + 1.$$

The normalized tails equal $\min\{1, \max\{0, x_i/(n-i+1) - t\}\}$, so they remain weakly decreasing. The first inequality holds since $b_2^{(t)} \le n-1$. Thus $b^{(t)} \in L_n$ and its first-coordinate difference is 1; it is an element of $\mathcal{H}_n$. For $i \ge 2$, the original inequalities give $0 \le x_i \le g(n-i+1)$, and the elementary truncation identity gives $\sum_t b_i^{(t)} = x_i$. The first coordinates sum to $x_2 + g = x_1$. This proves generation. Finally a vector with difference 1 cannot split into two nonzero vectors, since each has positive integral difference. □

The following principle is useful beyond the present problem.

Lemma 5.2 (Saturation from equal leading growth)**.** *Let $C \subseteq \mathbb{R}^n_{\ge 0}$ be a full-dimensional rational polyhedral cone and $L = C \cap \mathbb{Z}^n$. Let $A \subseteq k[y_1, \dots, y_n]$ be a subalgebra, with degree-lexicographic initial algebra. Suppose*

$$k[L] \subseteq \operatorname{in} A,$$

and, for the filtration by total degree at most N,

$$\dim_k A_{\le N} = cN^n + O(N^{n-1}), \qquad \#\{e \in L : |e| \le N\} = cN^n + O(N^{n-1}).$$

Then $\operatorname{in} A = k[L]$.

Proof. Gaussian elimination on the finite-dimensional vector space $A_{\le N}$ gives a basis with distinct leading monomials. Consequently its dimension is the number of monomials of degree at most N in $\operatorname{in} A$. For the reverse direction in this equality, a product of leading monomials is the leading monomial of the product of the underlying polynomials; degree-lexicographic order ensures the latter polynomial has the same total degree as that leading monomial.

Suppose that $y^u \in \operatorname{in} A$ with $u \notin C$. Choose one of the defining linear inequalities $f \ge 0$ for C that is violated by u, so $f(u) < 0$. Choose linearly independent integer vectors $v_1, \dots, v_n \in C$. Fix small positive constants κ, η such that

$$\kappa|u| + \eta \sum_i |v_i| < 1, \qquad \kappa f(u) + \eta \sum_i f(v_i) < 0.$$

This is possible by first choosing κ and then η. For all sufficiently large N, the vectors

$$\lfloor \kappa N \rfloor u + \sum_{i=1}^n m_i v_i, \qquad 0 \le m_i \le \lfloor \eta N \rfloor,$$

have total degree at most N, violate $f \ge 0$, and belong to the exponent semigroup of $\operatorname{in} A$. They are distinct because the v_i are independent. Thus, in addition to all the lattice points in C of degree at most N, the initial algebra contains at least $(\lfloor \eta N \rfloor + 1)^n$ other such monomials. This contradicts the equal leading growth constants. Hence no u outside C occurs. Since L contains every lattice point in C, the asserted equality follows. □

Lemma 5.3 (The two growth constants)**.** *For the Lecture Hall algebra and cone,*

$$\dim_k A_n(k)_{\le N} = \frac{2^n}{(2n)!} N^n + O(N^{n-1}),$$
$$\#\{e \in L_n : |e| \le N\} = \frac{2^n}{(2n)!} N^n + O(N^{n-1}).$$

Proof. The unique Dyck path of semilength $i-1$ that reaches height $i-1$ is the single mountain. Thus for $i \ge 2$

$$\ell_i = c_i y_i + q_i(y_1, \dots, y_{i-1}), \qquad c_i = \left(\prod_{h=1}^{i-1} \epsilon_h \right) \prod_{j=1}^{i-1} y_j^2 \ne 0. \tag{16}$$

Together with $\ell_1 = y_1$, this proves algebraic independence over every field: in a putative relation viewed as a polynomial in ℓ_i, the coefficient of the highest power of y_i is nonzero by induction. Each ℓ_i has total degree $2i-1$. Therefore the first count equals

$$\#\{a \in \mathbb{Z}^n_{\geq 0} : \sum_{i=1}^{n}(2i-1)a_i \leq N\} = \frac{N^n}{n!\prod_{i=1}^{n}(2i-1)} + O(N^{n-1}) = \frac{2^n}{(2n)!}N^n + O(N^{n-1}).$$

The estimate counts integer points in a weighted simplex. It follows, for example, by attaching a unit cube to each nonnegative integer point: the union lies between the weighted simplices of bounds N and $N + \sum_i(2i-1)$.

For the second count the cone is generated by the independent integer rays

$$v^{(j)} = (n, n-1, \ldots, n-j+1, 0, \ldots, 0), \qquad 1 \leq j \leq n.$$

If V is their column matrix, $|\det V| = n!$, and $w_j = |v^{(j)}| = j(2n-j+1)/2$. Hence

$$\mathrm{vol}\{x \in C_n : |x| \leq 1\} = \frac{|\det V|}{n!\prod_j w_j} = \frac{2^n}{(2n)!}, \qquad \prod_j w_j = \frac{(2n)!}{2^n}.$$

Integer-point count equals this volume times N^n, up to $O(N^{n-1})$. Here is an elementary justification: cover the dilated simplex by unit lattice cubes. Cubes wholly inside contribute their full volume and one lattice point under a fixed half-open convention. Only cubes intersecting the boundary can affect the difference. Each of its finitely many facets intersects $O(N^{n-1})$ cubes: solve its hyperplane equation for a coordinate with nonzero coefficient after fixing the other $n-1$ cube coordinates. There are $O(N^{n-1})$ such choices and a bounded number of remaining coordinate choices. This proves the required error estimate. □

6 The initial algebra and the Lecture Hall series

Proof of Theorem 1.1. Every ℓ_S belongs to $A_n(k)$ by its determinant definition. Theorem 1.2 and Proposition 4.1 show that its monic leading terms run through y^h, $h \in \mathcal{H}_n$. Lemma 5.1 therefore gives $k[L_n] \subseteq \mathrm{in}\, A_n(k)$. Lemmas 5.3 and 5.2 prove the reverse inclusion. The leading monomials of $\mathcal{B}_n$ generate this initial algebra, so $\mathcal{B}_n$ is a SAGBI basis by definition. Since the exponent vectors are distinct indecomposable elements of the semigroup, none can be removed. This also proves minimality. □

Corollary 6.1 (Bivariate Lecture Hall Theorem)**.** *As a formal power series,*

$$\sum_{e \in L_n} q_1^{e_1+e_3+\cdots} q_2^{e_2+e_4+\cdots} = \prod_{i=1}^{n} \frac{1}{1 - q_1^i q_2^{i-1}}.$$

Proof. Give odd-indexed y_j bidegree $(1,0)$ and even-indexed y_j bidegree $(0,1)$. Every adjacent weight $y_h y_{h+1}$ has bidegree $(1,1)$, so ℓ_i has bidegree $(i, i-1)$. Algebraic independence gives the product on the right as the Hilbert series of $A_n(k)$. Gaussian elimination within each finite-dimensional bihomogeneous component shows that taking initial terms preserves this Hilbert series. Theorem 1.1 identifies the initial algebra with the monomial algebra on the left. □

This is the classical theorem of Bousquet–Mélou and Eriksson, recalled in [1, Theorem 1.2]; the identity itself is not new. The proof here derives it from the initial-algebra result and elementary volume comparison, rather than using it to prove that result. In particular, the argument has no circular dependence on the Lecture Hall Theorem.

7 An integral basis and an explicit flat family

Set $A_n(\mathbb{Z}) = \mathbb{Z}[\ell_1, \dots, \ell_n]$.

Theorem 7.1 (Integral and deformation extensions)**.** *The set $\mathcal{B}_n$ is a monic SAGBI basis over $\mathbb{Z}$ and*

$$\operatorname{in} A_n(\mathbb{Z}) = \mathbb{Z}[L_n].$$

There is an explicit finitely generated $\mathbb{Z}[t]$-subalgebra $\mathcal{R}_n \subseteq \mathbb{Z}[t, y_1, \dots, y_n]$, free as a $\mathbb{Z}[t]$-module, with

$$\mathcal{R}_n/(t-1)\mathcal{R}_n \cong A_n(\mathbb{Z}), \qquad \mathcal{R}_n/t\mathcal{R}_n \cong \mathbb{Z}[L_n].$$

The same construction remains free, hence flat, after extension to any field. Thus it supplies the toric deformation in all characteristics.

Proof. Embed the integral algebra in $A_n(\mathbb{Q})$. The leading exponent of any nonzero integral polynomial in the algebra belongs to L_n by Theorem 1.1. For each $e \in L_n$, choose a decomposition $e = h_1 + \cdots + h_a$ into elements of $\mathcal{H}_n$, for example the one in Lemma 5.1. Use Proposition 4.1 to select $S(h_j)$ and put

$$F_e = \prod_{j=1}^{a} \ell_{S(h_j)}, \qquad F_0 = 1.$$

It has leading monomial y^e with coefficient 1. Repeatedly subtracting an integer multiple of the corresponding F_e from a polynomial cancels its leading term and strictly decreases the leading monomial. The process terminates: total degrees do not increase, and there are finitely many monomials in a fixed degree bound. It uses no division. Thus the F_e span $A_n(\mathbb{Z})$; their distinct leading monomials also make them linearly independent over $\mathbb{Z}$. This proves the integral SAGBI assertion.

For an explicit weight vector, let $D_n = n(n+1)/2$ and set

$$W_j = (D_n + 1)^{n-j} \qquad (1 \le j \le n).$$

Every generator ℓ_S has total degree at most D_n. Indeed for $r = |S| > 0$ the degree is $2\sum_{s\in S} s + 2r - r^2 \le r(2n+1-2r) \le D_n$; the last inequality follows because $D_n - r(2n+1-2r) = (n-2r)(n-2r+1)/2 \ge 0$. The empty subset has degree 1. These weights make the lexicographic leading monomial of each generator its unique greatest-weight monomial. At the first differing exponent coordinate, its positive contribution is at least W_j, whereas all possible negative contributions from later coordinates are at most $D_n W_{j+1} < W_j$.

For $h \in \mathcal{H}_n$ define

$$\widetilde{\ell}_h(t, y) = t^{W\cdot h}\ell_{S(h)}(t^{-W_1}y_1, \dots, t^{-W_n}y_n), \quad \mathcal{R}_n = \mathbb{Z}[t, \widetilde{\ell}_h : h \in \mathcal{H}_n].$$

The displayed expressions are polynomials: every occurring monomial has weight at most $W \cdot h$. For each $e \in L_n$ similarly put

$$\widetilde{F}_e = t^{W\cdot e}F_e(t^{-W_1}y_1, \dots, t^{-W_n}y_n).$$

It is a product of the indicated generators.

Each F_e has unique highest-weight monomial y^e. In the unique integral basis expansion of a polynomial, the largest weight among the occurring F_e cannot cancel: their highest-weight monomials are distinct. Since $F_eF_f - F_{e+f}$ has weight strictly below $W \cdot (e+f)$, it follows that

$$F_eF_f = F_{e+f} + \sum_{W\cdot g < W\cdot(e+f)} c_g F_g, \qquad c_g \in \mathbb{Z}.$$

Homogenizing yields

$$\widetilde{F}_e\widetilde{F}_f = \widetilde{F}_{e+f} + \sum_g c_g t^{W\cdot(e+f)-W\cdot g}\widetilde{F}_g.$$

Thus the $\mathbb{Z}[t]$-span of the $\widetilde{F}_e$ is closed under multiplication and is exactly $\mathcal{R}_n$. Their distinct leading y-monomials, each with coefficient 1, show that this span is free over $\mathbb{Z}[t]$. Modulo t the displayed multiplication becomes $\widetilde{F}_e\widetilde{F}_f = \widetilde{F}_{e+f}$, giving $\mathbb{Z}[L_n]$. At $t = 1$ the basis specializes to the basis F_e of $A_n(\mathbb{Z})$. After inverting t, the invertible change of variables $y_j \mapsto t^{-W_j}y_j$ and invertible scalar factors on the generators give $\mathcal{R}_n[t^{-1}] \cong A_n(\mathbb{Z})[t, t^{-1}]$. Base change preserves this explicit free basis and both fiber descriptions. □

8 Finite records and scope

The accompanying data contain finite exact records with the scopes below. The full extension suite and the earlier polynomiality suite were not rerun for this edition. The recorded regeneration of the extension data was completed in separate stages after an interruption; original process logs are not included. Stored counts and hashes do not independently certify that execution history.

The verifier reconstructs the finite tests summarized below. Its input is the Dyck-path definition and the original Toeplitz minors; the leading-term formula is used only as the expected answer.

Check	Exact scope and output
Original flag minors	All $2^{10} = 1024$ subsets of [10], including the empty subset, in 11 variables. All leading exponents and coefficients agree. The expanded polynomials have 884566 terms in total; the largest has 5427 terms.
Generic Schur determinants	All 195 partitions of sizes 0 through 11, computed by exact Bareiss determinants of independently generated moments.
Bijection and inverse	Every subset for each $1 \le n \le 16$: 65535 formula checks. These check the combinatorial formulas, not symbolic determinants in dimension 16.
Lattice and Hilbert counts	Direct enumeration in each $1 \le n \le 8$ through total degree 40: 24485 vectors overall. Bigraded counts agree with the weighted polynomial algebra, and each layer decomposition is checked.
Full small polynomials	All 128 minors for subsets of [7] are recorded as complete sparse coefficient lists in 8 variables.

The directory `code/` contains the executable verifier and comparison script. The directory `data/` contains all expected outputs, including leading exponents, partition profiles, term counts, and canonical polynomial hashes for the larger symbolic tests. Hash formatting is documented in the top-level README. Running the verifier regenerates the six mathematical data files. The earlier polynomiality manuscript and its separate verification files remain at the top level of this research package.

The proofs, not the finite tests, establish the all-n statements. The work does not claim that a numerical specialization retains algebraic independence or a SAGBI basis, and does not address uniqueness of the full defining system at vanishing divisors. Its algebra statements concern independent indeterminates. The generic Schur lemma, the explicit bijection, and the all-dimensional conclusions are proved above; they are not listed as conjectures or as extrapolations. A limited literature check located the original question and its publication record but does not certify priority for these arguments. Independent mathematical review remains appropriate.

AI assistance. This extension was developed with AI assistance. All mathematical reasoning needed for the stated results is presented here, and the exact verification code and data are provided for independent review.

A Identification with the defining Toeplitz equations

This appendix reproduces the foundation established in [2]. It is included for completeness, not claimed as a new extension. For a sequence P, let

$$M(P)_{r,c} = \begin{cases} -P_{c-r+1} & c \geq r, \\ 0 & c < r. \end{cases}$$

Define the square minors and target monomials by

$$E_i(P) = -\det M(P)_{\{1,\dots,\lceil i/2\rceil\},\{\lfloor i/2\rfloor+1,\dots,i\}}, \qquad T_i = \prod_{j=1}^{i} y_j^{i+1-j}.$$

Lemma A.1 (Hankel products)**.** *For independent Dyck weights and every $m \geq 1$,*

$$H_m^{(0)} := \det[\mu_{i+j}]_{0\leq i,j<m} = \prod_{h=1}^{m-1} (\alpha_{2h-1}\alpha_{2h})^{m-h},$$
$$H_m^{(1)} := \det[\mu_{i+j+1}]_{0\leq i,j<m} = \alpha_1^m \prod_{h=1}^{m-1} (\alpha_{2h}\alpha_{2h+1})^{m-h}.$$

Proof. Pair consecutive steps of a Dyck path. This gives a Motzkin path with up weight 1, down weight $\lambda_h = \alpha_{2h-1}\alpha_{2h}$, and horizontal weights $b_0 = \alpha_1$ and $b_h = \alpha_{2h} + \alpha_{2h+1}$ for $h \geq 1$. Let J be the tridiagonal matrix on nonnegative indices, with diagonal b_h, superdiagonal 1, and subdiagonal λ_h. Then $\mu_j = (J^j)_{0,0}$ by path enumeration. Put $d_0^* = 1$, $d_h^* = \lambda_1 \cdots \lambda_h$, and $D = \operatorname{diag}(d_0^*, d_1^*, \dots)$. Direct multiplication gives $JD = DJ^{\mathsf{T}}$, hence $(J^j)_{h,0} = d_h^*(J^j)_{0,h}$.

Fix m and write $L_{ih} = (J^i)_{0,h}$ for $0 \leq i, h < m$. This is unit lower triangular. If J_m, D_m are the leading blocks, path length bounds give the exact factorizations

$$[\mu_{i+j}]_{i,j<m} = LD_mL^{\mathsf{T}}, \qquad [\mu_{i+j+1}]_{i,j<m} = LJ_mD_mL^{\mathsf{T}}.$$

For the second identity split a path after i and then one more step; the two interface heights are at most i and j, respectively, so no term outside these finite blocks occurs. The first determinant is now $\prod_{h=0}^{m-1} d_h^*$.

For $\delta_m = \det J_m$, tridiagonal expansion gives

$$\delta_0 = 1, \quad \delta_1 = \alpha_1, \quad \delta_m = (\alpha_{2m-2} + \alpha_{2m-1})\delta_{m-1} - \alpha_{2m-3}\alpha_{2m-2}\delta_{m-2}.$$

Induction cancels the two terms involving α_{2m-2} and yields $\delta_m = \prod_{h=1}^{m} \alpha_{2h-1}$. Taking the second determinant and rearranging factors gives the shifted product. No division or sign assumption is used. □

Proposition A.2 (The moment identification)**.** *The polynomials in* (1) *satisfy $E_i(\ell) = T_i$ for every $i \geq 1$. Over the fraction field of the independent y_j, they are the unique sequence satisfying those equations.*

Proof. Put $s_m = (-1)^{1+m(m+1)/2}$. Row reversal, row signs, and the outer minus in the Toeplitz minors give

$$E_{2m-1}(\ell) = s_m y_1^m H_m^{(0)}, \qquad E_{2m}(\ell) = s_m y_1^m H_m^{(1)}.$$

Substitute (1) in Lemma A.1. The exponents of y_j in the two products, after multiplication by y_1^m, are respectively $2m-j$ and $2m+1-j$ on their supports. Their common sign, for $m \geq 2$, is

$$(-1)^{\sum_{h=2}^{m-1}(m-h)} = (-1)^{(m-2)(m-1)/2} = s_m.$$

For $m = 1$ both signs are 1 directly. This proves all the equations.

For uniqueness, $E_1 = P_1$ and $E_2 = P_2$. For $i \geq 3$ the entry $-P_i$ occurs only in the first row and last column of its square minor. If $k = \lceil i/2 \rceil$, its coefficient in E_i is $(-1)^k E_{i-2}(P)$, by cofactor expansion and shifting indices. After the previous equations have been solved this coefficient is $(-1)^k T_{i-2} \neq 0$. Successive solution in the fraction field gives a unique new P_i each time. This identifies the sequence with the one in the cited polynomiality note and source. The argument does not assert stepwise uniqueness after a specialization that makes these coefficients vanish. □

References

[1] Lukas Katthän, *The Lecture Hall Cone as a toric deformation*, arXiv:1809.01377v1 (2018), especially Definitions 3.1 and 4.1, Conjecture 4.2, and Problem 5.1. `https://arxiv.org/abs/1809.01377v1`. Published version: *Algebraic and Geometric Combinatorics on Lattice Polytopes*, World Scientific (2019), 242–253, `https://doi.org/10.1142/9789811200489_0015`. The theorem numbering used throughout this report is the arXiv numbering.

[2] *Polynomiality of the Lecture Hall sequence via signed Dyck paths*, version 0.1.1, October 13, 2026. `https://github.com/Aurantium-47/lecture-hall-polynomiality/releases/tag/v0.1.1`. The moment identification is reproduced in Appendix A.

[3] Alan D. Sokal, *A simple algorithm for expanding a power series as a continued fraction*, Expositiones Mathematicae **41** (2023), 245–287. `https://arxiv.org/abs/2206.15434`. `https://doi.org/10.1016/j.exmath.2022.12.001`.

[4] Philippe Flajolet, *Combinatorial aspects of continued fractions*, Discrete Mathematics **32** (1980), 125–161. `https://algo.inria.fr/flajolet/Publications/Flajolet80b.pdf`.